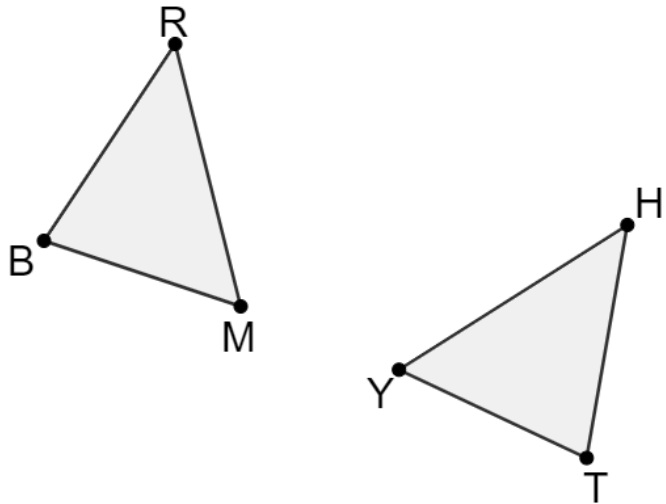


Triangles BRM and YHT are shown. Determine if the given information is sufficient to conclude that the triangles are congruent. If so, name the congruence theorem or postulate that shows congruence.



Given Information	Are They Congruent?	Theorem or Postulate Used
1. $\overline{MR} \cong \overline{YH}$, $\angle HYT \cong \angle BMR$, $\angle BRM \cong \angle THY$	Yes	ASA
2. $\overline{MB} \cong \overline{YT}$, $\overline{MR} \cong \overline{HY}$, $\overline{BR} \cong \overline{TH}$	Yes	SSS
3. $\overline{MB} \cong \overline{YT}$, $\angle YTH \cong \angle MBR$, $\angle BRM \cong \angle THY$	Yes	AAS

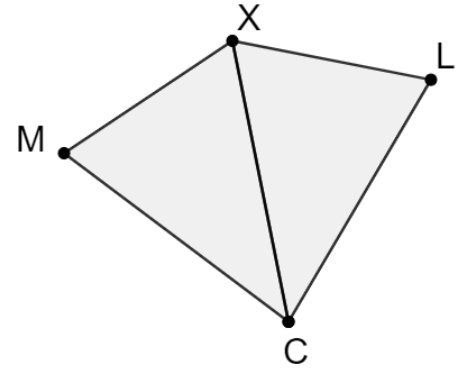
4. Quadrilateral $CLXM$ is shown where $\overline{MC} \cong \overline{LC}$ and $\angle MCX \cong \angle XCL$.

Complete the paragraph proof.

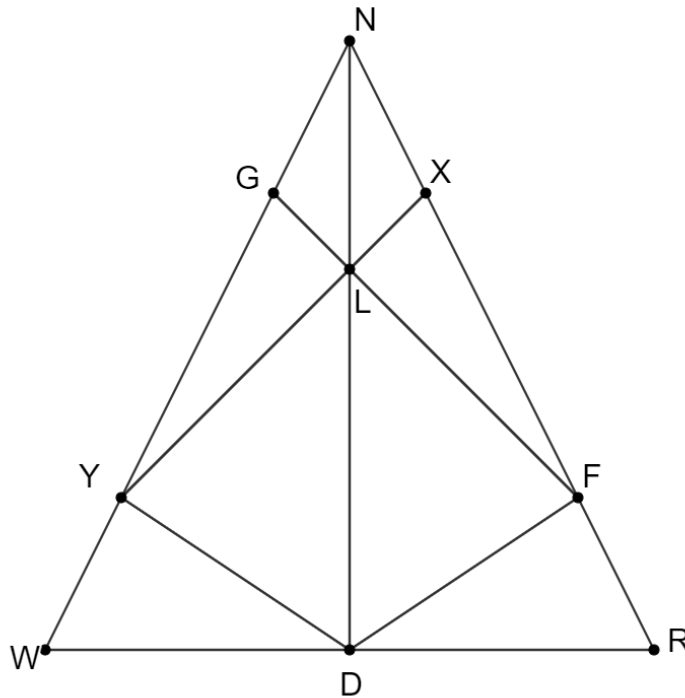
It is given that $\overline{MC} \cong \overline{LC}$ and $\angle MCX \cong \angle XCL$.

By the reflexive property, $\overline{XC} \cong \overline{XC}$. Therefore,

$\triangle MCX \cong \triangle LCX$ by **SAS**.



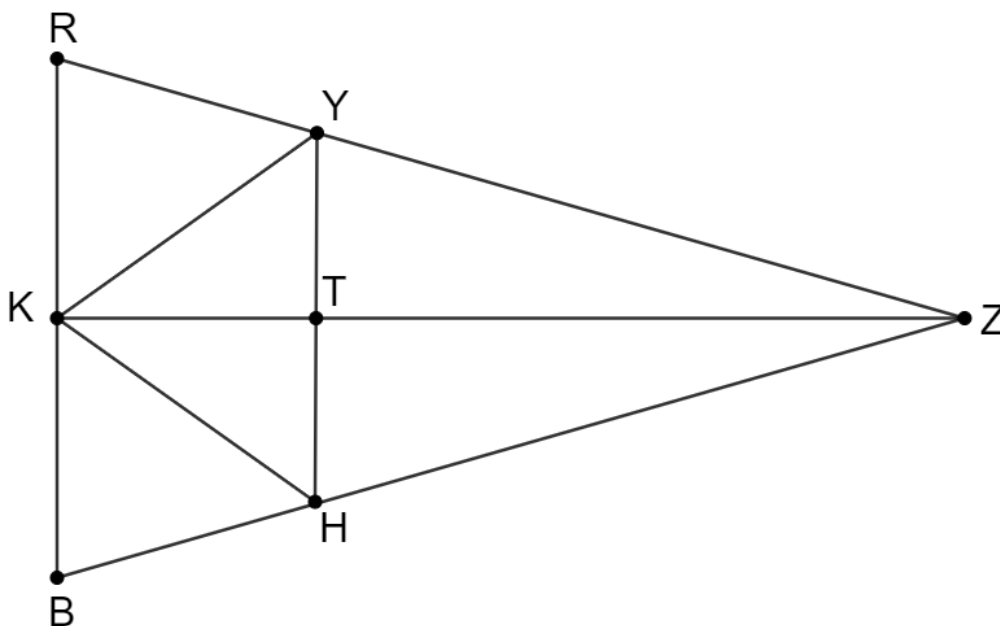
5. Triangle NRW is shown.



Complete each statement.

- If $\triangle FLD \cong \triangle YLD$ then $\angle DFL \cong \angle \textcolor{red}{DYL}$.
- If $\overline{ND} \perp \overline{WR}$ and $\overline{ND} \cong \overline{ND}$, the information needed to show that $\triangle NDW \cong \triangle NDR$ using HL is $\overline{NW} \cong \overline{NR}$.

6. Triangle RZB is shown.



Given: $\Delta KYT \cong \Delta KHT$

Prove: $\Delta YKZ \cong \Delta HKZ$

An incomplete proof is shown.

Statement	Reason
1. $\Delta KYT \cong \Delta KHT$	1. Given
2. $\overline{KY} \cong \overline{KH}$	2.
3.	3. Corresponding Parts of Congruent Triangles are Congruent (CPCTC)
4.	4. Reflexive Property
5. $\Delta YKZ \cong \Delta HKZ$	5.

What is the reason for step 5?

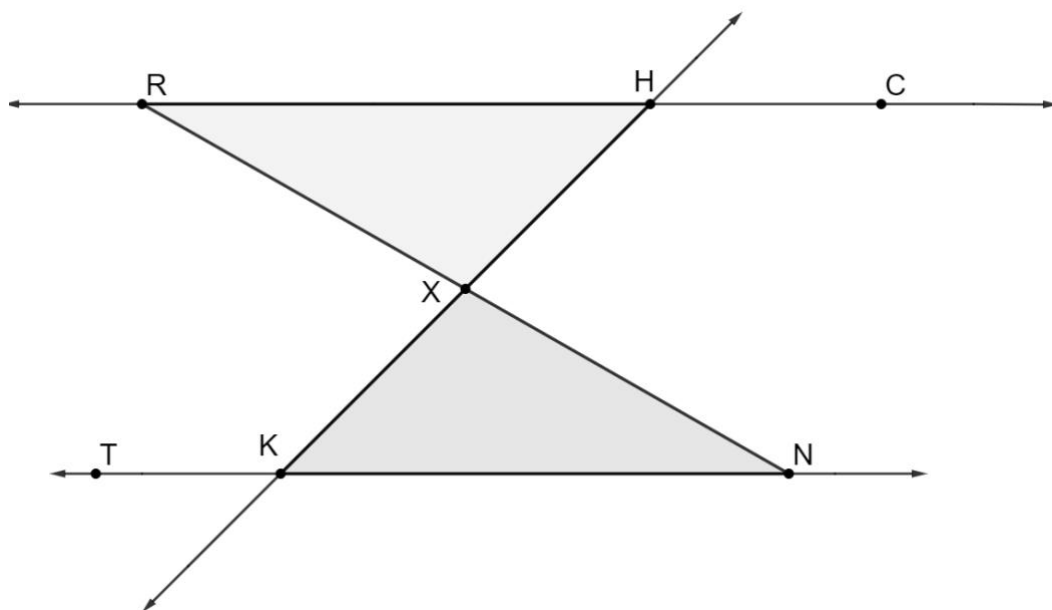
Reason for step 5	SAS
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7. Parallel lines $\overleftrightarrow{RC}$ and $\overleftrightarrow{TN}$ with transversal $\overleftrightarrow{KH}$ is shown.

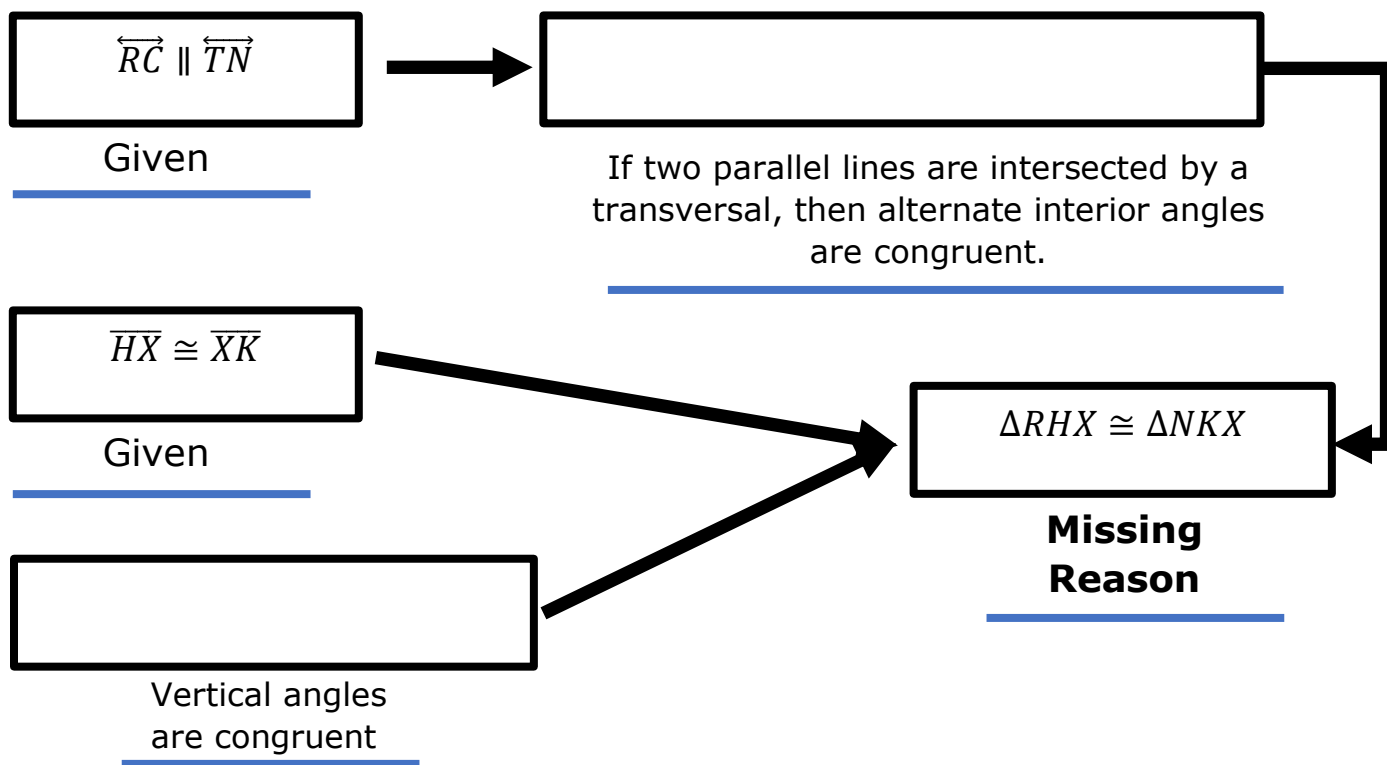
Given: $\overleftrightarrow{RC} \parallel \overleftrightarrow{TN}$,

$\overline{HX} \cong \overline{XK}$

Prove: $\triangle RHX \cong \triangle NKX$



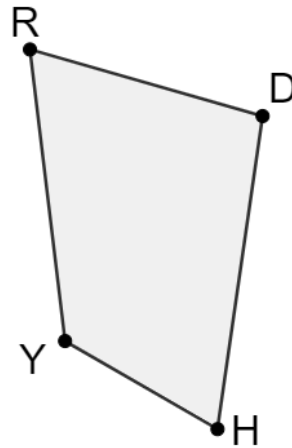
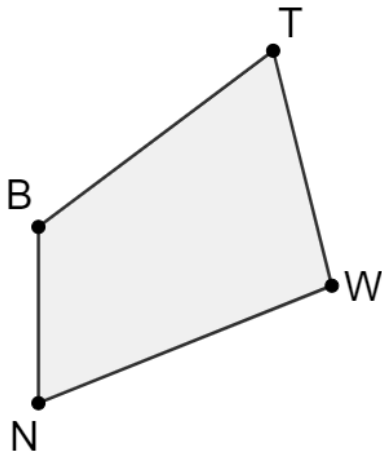
An incomplete flowchart proof is shown.



What is the missing reason in the flowchart proof?

Missing Reason	ASA
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8. Quadrilaterals $TWNB$ and $HYRD$ are shown.



It is given that $TWNB \cong RDHY$, $NW = 3x$, $RD = 5x - 19.6$, $RY = 2x + 5.6$, $HD = 5x - 14.4$, $TW = 3x - 5.2$, and $BN = x + 4.8$.

- a. Determine the value of x . Show your work.

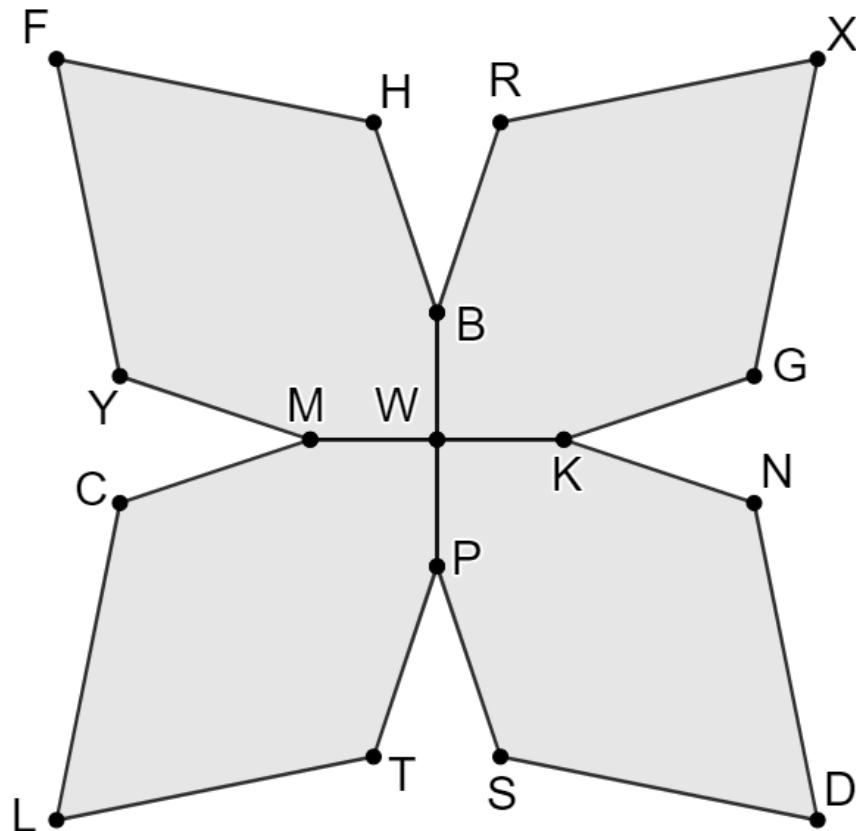
$x = 7.2$

- b. Determine the perimeter of quadrilateral $TWNB$. Show your work.

Perimeter	70
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9. A graphic artist created a design for a flower using four congruent hexagons as shown.

Given: $\overline{BP} \perp \overline{MK}$,
 $m\angle MYF = m\angle FHB$,
 $m\angle WMC = m\angle WPT$,
 $m\angle GXR = 66^\circ$,
 $m\angle DNK = (3x + 18)^\circ$, and
 $m\angle WBR = (5x - 8)^\circ$. The sum of the angles in $WKNDSP$ is 720° .



Find each. Show your work

Work

a. $m\angle TLC = 66^\circ$

b. $m\angle KGX = 120^\circ$